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Deep-Learning ASR for a Patient with Permanent Tracheostomy: A Case Study

Bachelor Thesis

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Abstract

This bachelor thesis presents a case study on the development of a personalized *automatic speech recognition* (ASR) system for a patient with severe speech impediment caused by a permanent tracheostomy and potential neurological damage following a debilitating car accident. We collect and curate a public dataset of their speech and fine-tune a deep-learning ASR model, Whisper, to recognize it. We utilize knowledge distillation from masked language model, thus achieving near healthy-voice accuracy for in-domain data as well as improving the publicly available smallest Whisper models for regular Czech speech. Additionally, we discuss limitations in real-world applicability of our system and propose several strategies how to overcome them. With these strategies, we achieve performance that is considered helpful by our patient.

Kurzfassung

In dieser Bachelorarbeit wird eine Fallstudie über die Entwicklung eines personalisierten Spracherkennungssystems (ASR) für einen Patienten mit schwerer Sprachbehinderung aufgrund eines permanenten Luftröhrenschnittes und möglicher neurologischer Schäden nach einem schweren Autounfall vorgestellt. Wir sammeln und kuratieren einen öffentlichen Datensatz ihrer Sprache und stimmen ein Deep-Learning-ASR-Modell, Whisper, darauf ab, sie zu erkennen. Wir nutzen die *knowledge distillation* aus *masked language model* und erreichen so eine annähernde Erkennungsgenauigkeit von gesunder Stimme für In-Domain-Daten sowie eine Verbesserung der öffentlich verfügbaren kleinsten Whisper-Modelle für normale tschechische Sprache. Darüber hinaus diskutieren wir die Grenzen der Anwendbarkeit unseres Systems in der realen Welt und schlagen verschiedene Strategien vor, um diese zu überwinden. Mit diesen Strategien erreichen wir eine Leistung, die von unseren Patienten als hilfreich empfunden wird.

Contents

Abbrevations and Notation

AAC	Alternative and Augmentative Communication
ASR	Automatic Speech Recognition
CER	Character Error Rate
CNN	Convolutional Neural Network
CTC	Connectionist Temporal Classification
DL	Deep Learning
DNN	Deep Neural Network
HMM	Hidden Markov Model
KD	Knowledge Distillation
LLM	Large Language Model
LSTM	Long Short-Term Memory
ML	Machine Learning
MLM	Masked Language Model
TSNE	t-Distributed Stochastic Neighbor Embedding
PCA	Principal Component Analysis
RNN	Recurrent Neural Network
SNR	Signal-to-Noise Ratio
WER	Word Error Rate

1 Introduction

This thesis is a case study of a Czech teenage girl ('Tali') who was struck by a car at a preschool age. The accident left her quadriplegic and she requires a permanent tracheostomy to help her breathing. The adverse effect of the tracheostomy is, however, a severe speech impediment which is, possibly, further amplified by the impaired motorics resulting from her spine injuries. Although the family relatives or experienced assistants can learn to understand her speech, for a general Czech speaker, Tali's speech is unintelligible. We discuss her condition in more detail in Chapter 2.

Figure 1.1: Face-swapped images of Tali. The picture on the left highlights the stoma on her throat. The picture on the right depicts her communication with experienced assistants

In this thesis we describe a development of a deep-learning ASR system that would help Tali to verbally communicate even with untrained conversational partners. For this purpose we organize over one hundred speaking sessions to collect 24 hours of annotated recordings of her speech. Afterwards, we fine-tune Whisper, ASR transformer by OpenAI, on this data. For smaller Whisper models, we achieve near healthy-voice accuracy for the data collected in the standard sessions.

We also discuss the results of our system on the real-world conversations and propose several strategies how tackle the challenges caused by the domain shift. With these strategies, we achieve 0.41% transcription accuracy that is already considered useful by Tali and her caretakers.

All our codes are available on GitHub[1].

1.1 Contribution

The key contributions of this work are:

- Creation and curation of a public dataset of Tali's speech that can be used also by other researchers and assistive technology developers.
- Demonstrating that KD from MLM into Whisper can improve the ASR quality, especially when the Whisper model is relatively small. As a part of this contribution, we also create and publish a Whisper Tiny trained this way that achieves the WER at least 6% better than the previously available models.
- Achieveing near healthy-voice ASR performance on the in-domain data with smaller Whisper models.
- Discussing challenges that come with deployment of the system in the real world and providing and testing strategies how to handle them.

1.2 Outline

The rest of this thesis is structured as follows: In Chapter 2 we provide a bit more detail on the causes of Tali's speech impediments, we also organize a small experiment to estimate the intelligibility of her speech to a general Czech speaker. In Chapter 3 we

[1] https://github.com/Hobit2002/TracheoSpeech_ASR

explore ASR methods while giving special focus to state-of-the-art DL models DeepSpeech, Wave2Vec and Whisper. Furthermore, we investigate the contemporary assistive tools for people with severe speech impediments ranging from classical and widely used low-tech tools to recent and more experimental methods. In Chapter 4 we describe in detail the data collection process, especially the *artificial conversations*, the predominant type of our recording sessions. In Chapter 5 we analyze Tali's speech, both quantitatively and qualitatively, and compare it to the standard one. In Chapter 6 we provide details on how we trained our models, namely the pre-training steps and the KD from MLM. In Chapter 7 we explain our data split and provide the results of different models on different datasets. In Chapter 8 we discuss the remaining challenges in deploying the model and present the decoding strategies with which we achieve practically applicable results. In Chapter 9 we summarize our results and provide possible directions of the future work.

2 Tali's condition

We find it appropriate to briefly summarize the available information on Tali's condition[1] as it appears to be very specific and deviating significantly from standard descriptions of tracheostomy and other common speech impediments.

Tracheostomy is a surgical procedure intended to help people with severe breathing difficulties such as mechanically blocked upper airways or respiratory failure. As we do not have access to Tali's medical records, we do not know, why she needed the procedure. During the surgery a stoma is created on the trachea of the patient and a plastic tube is inserted into it. Typically, the tracheostomy takes place only for a several days/weeks and the stoma closes naturally several days after the tube removal [49].

Figure 2.1: Tracheostomy as described in [51] - in Tali's case, the tube is not present

In this form, the tracheostomy also completely prevents patients from speaking as the air exhaled from the lungs leaves the throat via tube before ever reaching the larynx where the speech is produced. This problem however has a standard solution in the form of speaking valves that are attached to the tube and close it during exhalation, so that the air has to go through the larynx [51]. The valve users are capable of well understandable speech [9].

[1]Which is limited as we did not get access to her medical documentation.

Clearly, Tali's condition is different in many aspects - at this time she has lived with the stoma for many years, she does not use the tube and she is capable of speech production. Therefore in her case, *long-term tube-free tracheostomy* [24] or *persisten tracheal stoma* [54] might be more medically accurate terms. However, as this is not a medical project and as Tali's most visible speech impediment originates from having a stoma in the trachea, we find it appropriate to use the short term *tracheostomy*.

Furthermore, impaired motor control or general muscle weakness resulting from the neurological damage might be hindering Tali's speech as well.

2.1 Intelligibility Estimation

In order to estimate the intelligibility of Tali's speech by people not accustomed to it, we asked eight respondents to listen to twenty five segments of Tali's speech (twenty of them being sampled from a real-world conversation and five from an artificial one) and to try to transcribe it. We played the audio with high volume, the errors below therefore cannot be attributed purely to the speech being too quiet. We played every sample exactly once. The time for writing the transcriptions was not limited. One of the respondents already had experience with Tali as they assisted with the recording day (see Section 4.1.3). We provide their results separately.

	At least one	**All**
All respondents	0.128	0.031
Respondents excluding the assistant	0.103	0.023
The assistant	0.28	0.08

Table 2.1: The relative frequencies of at least one/all words being correctly recognized by a participant for a sample

Although our experiment used only a small number of both participants and listening samples, it demonstrates that Tali's speech is indeed not understandable for an untrained listener. In general, the words recognized by the respondents were rather short. Notably, the only sample where the respondents other than the assistant were able to recognize all the words actually contained only a single word which was *Ne* (*No* in English). Furthermore, the respondents demonstrated considerable confusion and helplessness during the experiment and even fatigue in its later part.

Both the quantitative and the qualitative observations indicate that an ASR system could be very beneficial for Tali.

3 Related work

As to our knowledge, scientific literature offers no precedence of using state-of-the-art deep-learning ASR methods as an assistive tool for people with permanent tracheostomy. The related works for this thesis therefore come from two separate fields with only limited exchange of ideas: DL ASR and assistive technologies for people with speech disabilities. To account for their limited overlap, we dedicate a separate section to each of those topics.

3.1 Deep-learning ASR

As the ways of uttering a sentence largely vary among and even within individual speakers, ASR has been from its very begginings heavily reliant on statistical and ML methods. This is evidenced by the reasonings of Davids, et.al, inventors of the first ASR system (speaker-dependent spoken digit recognition) in 1950s: *Since design of any successful recognition circuit demands a quantitative knowledge of an inherently variable speech signal, any useful description of this signal must be expressed in statistical terms.* [23]

The authors of the aforementioned work computed frequency statistics of individual digits (as pronounced by a single speaker) and then hardwired those into their ASR circuit. This template-based approach has been prevalent in the field also for the following decades [39]. The shift to ML approaches came in 1970s and 1980s [37]. At that point, HMMs became the standard algorithm for ASR [27] and have been widely used in practice till today.

The 21st century brought more focus on DNNs. In 2008, Graves, et.al proposed CTC [30], an RNN-based ASR system, that slighlty outperformed both HMM and hybrid HMM-DNN systems on a dataset of phonetically segmented and annotated English speech. Critical challenge of this approach was mapping the RNN predictions for individual audio frames to a sequence of discrete labels that had a (much) shorter length. To mitigate this ill-alignment between the RNN's outputs and targets, the authors design *CTC loss* function that uses a forward-backward algorithm to compute the likelihood of a valid label sequence given the model's predictions. Their reasoning is built upon a simplifying assumption that the model's outputs are mutually conditionally independent given the internal state of the network. For the details we refer an enthusiastic reader to the original paper.

In the following years the researchers aimed to develop *end-to-end* neural ASR methods that would mitigate shortcomings of CTC (i.e. its conditional independence assumption and non-trivial training and inference algorithms). A particularly important innovations were *sequence-to-sequence learning* [56] and *attention* [12].

Sequence-to-sequence learning removes the necessity of handling the alignment between the input sequence and targets as the input processing is done by a dedicated RNN, the *encoder*, and results into an embedding that is then processed by another RNN, the *decoder*, that outputs the final transcription. This enables for *end-to-end* ASR as has been utilized by Sak, et.al in the Recurrent Neural Aligner [52].

Sequence-to-sequence learning as proposed in the original paper, however, had the caveat of using a fixed-size hidden state which limited its capability of processing longer sequences. To overcome this problem, Bhandau, et.al [12] utilize the *attention* mechanism. Their architecture comprises an *alignment model* that learns how is each of the input tokens relevant for each of the output ones. We omit the computational details as *attention* mechanism has many variants and their rigorous survey would suffice for a standalone theis. As an example of an early use of *attention* for ASR, we refer an interested reader to *Listen, Attend and Spell* by Chan, et.al [18].

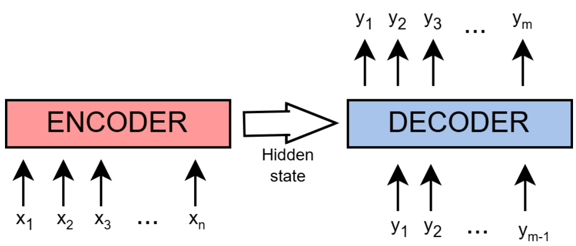

Figure 3.1: Sequence-to-sequence architecture

Sequence-to-sequence learning and *attention* are jointly used for ASR by Google in 2017 [20] and for the first time outperform the classical HMM approaches on a Google's dataset of 12 500 hours of spoken English with noisy background. Perhaps even more notably, they serve as core mechanisms of a general-purpose *transformer* [59] architecture that omits the recurrent component of the classical *sequence-to-sequence* architecture and thus allows highly efficient parallelization.

The current state-of-the-art models generally use the algorithms presented above and achieve low error rates thanks to large number of parameters. Below, we present the current open-source state-of-the-art models (i.e. those we considered for our own project) in detail.

3.1.1 DeepSpeech

Although the DeepSpeech [31] model dates back to 2014, it still achieves competitive performance for the Czech speech [3] (15% WER for Common Voice dataset [8]) among models that are sufficiently light-weight for real-time inference.

In the original paper, DeepSpeech takes spectrograms as inputs. It is a bidirectional RNN whose timesteps corresponds to temporal bins of the spectrogram. The original DeepSpeech consists of five layers. The first three process only the input of the current timestep and the fourth one is a bidirectional RNN. The last layer then again operates frame-wise. As opposed to the open-source design by Mozilla depicted in Figure 3.2, the original paper used vanilla RNN and not LSTM for the recurrent layer. The authors later introduced DeepSpeech 2 [7] that uses up to seven recurrent layers.

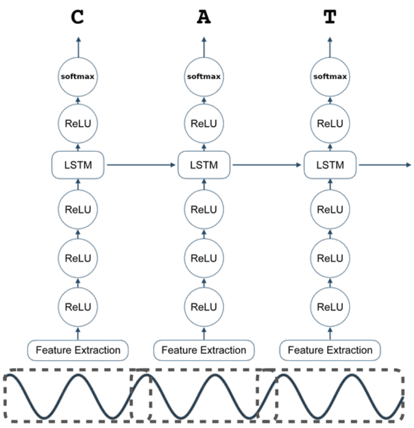

Figure 3.2: DeepSpeech architecture as implemented by Mozilla [1]

DeepSpeech is trained with *CTC loss function*. It is commonly used in combination with an n-gram language model that helps it to stick with exististing vocabulary of the language. Specifically, the transcription maximizes the following objective function:

$$Q(\text{trans.}) = \log P_{ASR}(\text{trans.}|\text{sound}) + \alpha \log P_{LM}(\text{trans.}) + \beta\, \text{word\_count}(\text{trans.}) \quad (3.1)$$

where $P_{ASR}(\text{trans.}|\text{sound})$ stands for transcription probability as estimated by the DeepSpeech and $P_{LM}(\text{trans.})$ for transcription probability as estimated by the n-gram. α and β are standard hyperparameters. The authors of the original paper maximized the objective with a beam-search of beam size 1000 - 8000. For the Czech Common Voice, the DeepSpeech achieved almost three times lower WER when integrated with n-gram (15% with vs 41% without) [3].

DeepSpeech has been implemented and provided as open-source under a public license by Mozilla [1]. The only difference from the original solution was the use of LSTM instead of RNN in the recurrent block. DeepSpeech is also provided by *Torchaudio* [2].

3.1.2 Wave2Vec

Wave2Vec models [11, 53], have been shown to not only outperform DeepSpeech 2.0 on the WSJ benchmark [28] (various speakers reading aloud articles from the Wall Street Journal) but they also perform very well on the Czech speech with Wave2Vec 2.0 achieving 4% WER with 90M parameters [41].

Both the original Wav2Vec [53] and the newer Wav2Vec 2.0 [11] utilize unsupervised pre-training. In the pre-training phase the contrastive loss is used and models learn to distinguish between audio segments fitting to the provided context and distractor segments. Specifically, the high-level unsupervised objective is to minimize the following error function:

$$\mathcal{L}(\text{cont.}, \text{sample\_true}) = -\log \frac{\text{sim}(\text{g}(\text{cont.}), \text{sample\_true})}{\sum_{\text{sample\_distract}} \text{sim}(\text{g}(\text{cont.}), \text{sample\_distract})} \quad (3.2)$$

where sim stands for a similarity function, cont. for the context provided to Wav2Vec as an input and g as the model itself. After pre-training of this kind, Wav2Vec models were trained using the CTC loss in a supervised manner. It was shown that after being pre-trained on vast amounts of unlabelled data (thousands of hours in magnitude), they need only a few hours of labelled data to achieve state-of-the-art performance on the respective domain and only a few minutes to achieve decent yet suboptimal results [11]. Similarly to Deep Speech, Wave2Vec models use a language model during inference to guide its predictions towards existing vocabulary [11, 53, 41].

Apart from the aforementioned principle, Wav2Vec [53] and Wav2Vec 2.0 [11] are largely different. Wav2Vec is a CNN that uses Mel spectrograms as inputs whereas Wav2Vec 2.0 combines convolutional layers with transformer ones and takes raw waveform as input. There are also notable differences in the implementation of the unsupervised pre-training, but for the precise details, we refer an interested reader to the respective papers.

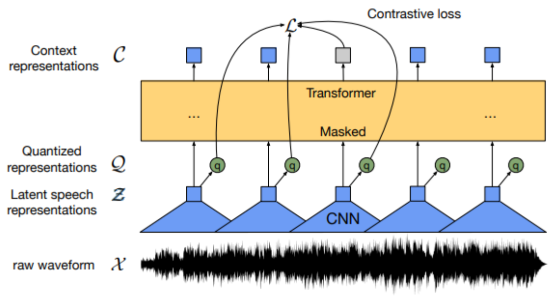

Figure 3.3: Wave2Vec 2.0 architecture - taken from the respective paper [11]

Pre-trained Wav2Vec is publically available via HuggingFace both for English [4] and for Czech [37].

3.1.3 Whisper

Whisper [50] is a relatively recent (2022) ASR model that exhibits strong zero-shot performance even on out-of-domain data. It is provided in seven versions of five different sizes of which the larger ones generalize to healthy Czech speech already quite well (large-v3 achieves zero-shot 9% WER on Czech Common Voice [5]). However, also for the smaller models, many fine-tuned versions exist on Hugging Face and claim high performance [58, 45, 40].

Whisper is a standard encoder-decoder transformer [59] taking Mel spectrograms as inputs. Byte pair encoding [26] is used for tokenizing the text. The spectrogram patches are processed by a small CNN with GELU activation function [33] before the positional sinusoidal embedding is added to them. The decoder uses learned positional embedding.

The main innovation behind the Whisper model was neither its architecture nor the training method, but rather the vast amount of data used for its training. Unlike the previously mentioned models that were trained on relatively limited data with human-validated labels such as Common Voice [8], Whisper used 860k hours of supposedly human-annotated yet not validated speech scrapped from the internet. Results of other ASR systems were filtered out from this data.

Whisper is provided in a dedicated Python library [5]. Furthermore, many fine-tuned versions are available via Hugging Face [58, 45, 40].

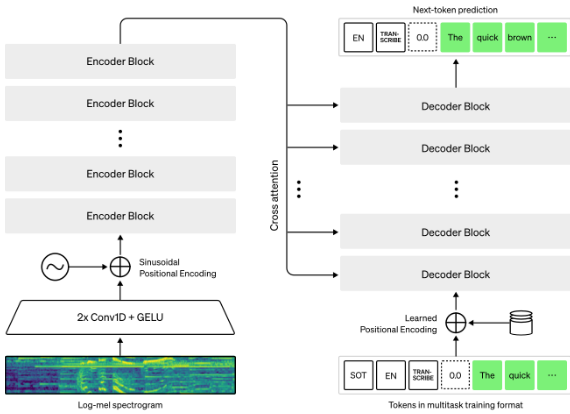

Figure 3.4: Whisper architecture - taken from the OpenAI site [5]

3.2 Alternative and Augmentative Communication

The term Alternative and Augmentative Communication (AAC) refers to techniques that help people with various communication impairments and disabilities to cope with their handicaps [14].

In the literature, AAC are categorized into the following two classes: low-tech and high-tech [47].

As *low-tech AAC* we typically understand approaches that do not require an external power source in order to work. Low-tech methods include writing the messages down, pointing to letters on a board, communication cards, sign languages, eye-blink codes, etc. [46]

High-tech AAC refers to more sophisticated approaches that typically utilize an electrical device and require a power-source to operate. Traditionally, they include tools that read aloud, what the user has written [60] or recognize a sign language [44]. ASR systems represent another typical high-tech approach [46].

The AAC ASR systems typically utilize the very same machine-learning algorithms as ASR systems for the healthy speech - HMMs [32] LSTMs [64], state-of-the-art ASR architectures as Wave2Vec 2.0 [34] or deep autoencoder for feature preprocessing [57]. Unfortunately, the relevance of these works was limited for our own project, because they largely focused on patients affected by dysarthria (neurological damage commonly caused by stroke or Parkinson's disease) and not the peculiar condition of Tali.

As the WER of AAC ASR systems is typically very high, a common approach is to limit the vocabulary to only a set of words. As an example, we refer to the work of Hawley, et.al. [32] who developed an AAC ASR system based on HMMs (VIVOCA) for seven people with severe dysarthria. The system recognizes only 10 - 50 words for every patient and these words are then used to navigate via a tree of predefined set of phrases like 'I would like a cup of coffee' or 'Could you open the window please?'. When the phrase is selected, an artificial voice reads it aloud. The system achieves over 90% accuracy for all but one patient. However, it received criticism from users for being too slow and for the vocabulary being limited too much.

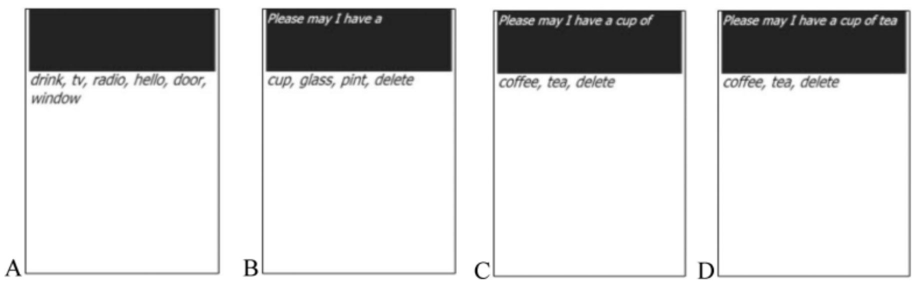

Figure 3.5: VIVOCA user interface [32]: on the highest level A the user chooses to ask for a drink, they are directed to level B where they specify that they want a cup of the drink, on level C they specify that drink in the cup should be tea. When the whole phrase is selected (D) the system reads it aloud.

Recently, brain2speech AAC systems were introduced. In particular, Card,et.al. [17] developed AAC for a patient with severe dysarthia caused by ALS that used neuroprosthetics to collect the data from the patient's brain. The system achieved 2.5% WER with only 16.6 hours of training data. For this purpose, the authors used four machine learning models to predict phonemes from the brain waves, to predict sentences from the phonemes, to choose the most appropriate sentence and to generate an artificial patient's voice saying the sentence out loud. We deem our own work to be relevant even in the context of these advancements as our method is non-invasive and less expensive.

4 Data Collection

The data collection was one of the most demanding tasks of this work. We performed over one hundred recording sessions and annotated over twenty hours of Tali's speech. This was possible only due to *artificial conversations*, a novel and efficient approach to data collection which we detail below along with other methods we utilized.

4.1 Data Collection Methods

We applied the following three methods for data collection:

- **Reading sessions**:
- **Artificial conversations**
- **Recording day**

Below, we examine each of them in detail.

4.1.1 Reading sessions

Reading sessions were the initial method used for data collection. Tali's task was to read a page of text of her choice (typically an article from a popular space magazine) and then send us both the scan of the page and a video recording of her reading.

During post-processing, we extracted text from the scan and split it to individual words by whitespaces. We then split the recording (MOV) to video (mp4) and audio (mp3). Both the original and the resulting audio had a sample rate of 44.1 kHz with 32 bits per sample. For audio we detected all the samples with amplitude over 800 PCM (determined experimentally) and stored the segments from 0.25s before the peak to 2s after the peak. Finally, we manually aligned the segments of audio to the pieces of text.

As Tali's voice is highly unintelligible and she has an unfortunate habit of enriching the provided scripts with her own modifications we found our labelling approach highly error-prone. To assess its quality, we developed a web interface [48] to verify and eventually correct our labelling. We then asked Tali's mother to perform these tasks for 2 reading sessions with approximately 250 samples in total. Based on this evaluation we estimated that only around 2-3% samples from the reading sessions are mislabelelled.

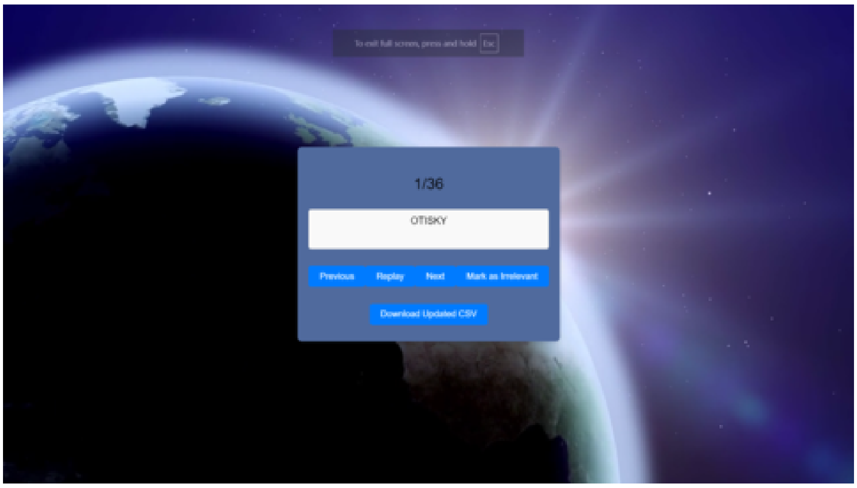

Figure 4.1: The interface for annotation verification

Although we found the error rate of 2-3% reasonably low, the reading sessions were nevertheless soon abandoned for the following reasons:

1. As post-processing a 15-minute session, typically requires around 90 minutes of highly concentrated work, it was unfeasible to collect the necessary 20 hours of data this way.

2. The vocabulary used in the space magazines is not representative of our target domain - real-world conversations.

3. The pace, tone and other non-verbal aspects of speech probably also differ significantly between reading and real-world conversations.

4. The Tali and her family had issues with scheduling the sessions and sending us their recordings sufficiently often.

4.1.2 Artificial conversations

The predominant type of recording sessions were *artificial conversations*. For those, we provided Tali with an audio file containing a dialogue of characters A and B. Her task was to repeat the lines said by character A. She recorded herself during the procedure and when finished, she sent us the video recording.

Figure 4.2: Artificial conversations - the upper line depicts the structure of the audio file prepared by us. The lower line displays at what moments we expected Tali to speak.

The process of generating the artificial conversations was loosely inspired by RefGPT [63], a method proposed for generation of truthful dialogues using LLMs. We prompted ChatGPT (gpt-4o-mini) [36] with the beginning of the dialogue, 1-4 sentences on its high-level topics, detail instructions on the manner in which the characters should speak and occasionally also a source article on the topic of the conversation. The result was split into lines said by character A and lines said by character B. Both groups were then turned to audio using Google Text-to-Speech [21]. The lines by character B furthermore went through voice conversion to make them distinguishable from the lines of character A. For the voice conversion we typically used the service Voice.ai [61], but occasionally we opted for Seed-VC [42] to enrich the conversations with a voice of publicly known person such as but not limited to Elon Musk. Finally, we merged the audio into one dialogue file that included also an intro and outro song excerpt (generated by Suno.ai [55]) as well as a recap of instructions. When creating the audio file, we also saved temporal coordinates of the silence segments during which we expected Tali to speak together with the text of corresponding lines. Those were later essential for processing the recordings efficiently.

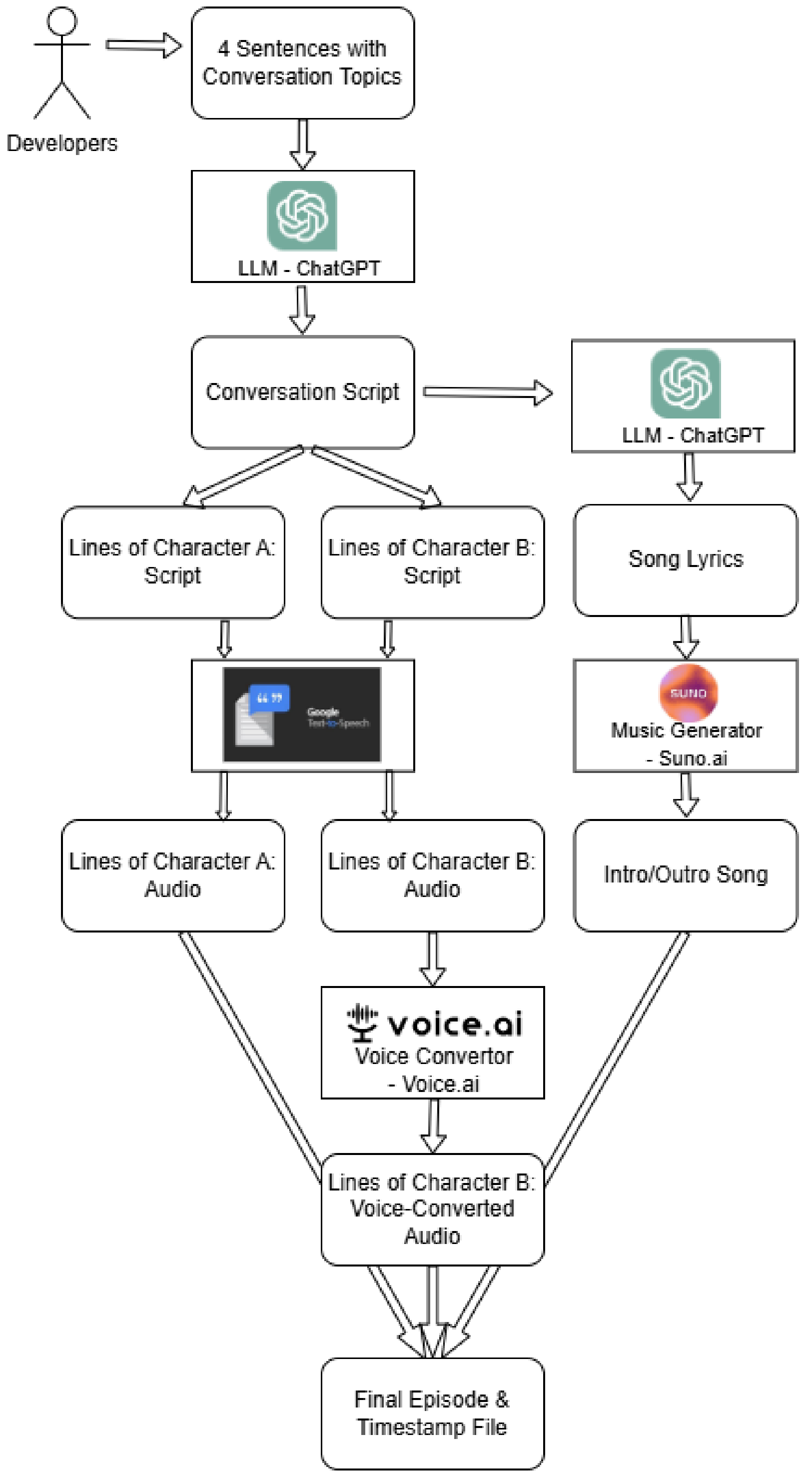

Figure 4.3: Artificial conversations - scheme of the generation process

As already mentioned, we provided Tali with 10 seconds of silence to repeat every A's line. However we later included also another 20-second silent segment after every 40 dialogue lines (by both characters). On these occasions, Tali had the opportunity to freely comment on the dialogue from her own perspective. Although we did not process and store these segments later, they helped Tali to vent her need for conversational agency which otherwise interfered with precisely repeating the prescribed lines.

In order to make the artificial conversations both engaging and representative of Tali's real-world vocabulary, we dedicated most of the episodes to topics such as Tali's friends, home city and hobbies. We also made episodes on various everyday topics such as school or family relationships. We also kept the lines short to mirror her real-world concise manner of speech.

Tomka: I have another idea.
Shalina: What idea?
Tomka: I want to give a hundred parrots as a gift.
Shalina: That's a big number! Why parrots?
Tomka: Martin ***** likes birds.
Shalina: That's right, he loves birds.
Tomka: I think they'll make his family happy.
Shalina: Where are you going to get them?
Tomka: We have them at the zoo.
Shalina: Who will give them to you?
Tomka: I want all 100 of them.
Shalina: What will you do with the old ones?
Tomka: I'll choose them carefully.
Shalina: What kind of parrots do you want?
Tomka: Mostly colourful and intelligent.
Shalina: That sounds great!
Tomka: Maybe even a few rare ones.
Shalina: Do you know when to donate yet?
Tomka: For Martin's Day, maybe.

Figure 4.4: Conversation Excerpt. Tomka is the character A. Shalina is the character B. We masked out the family name of Tali's real-world friend Martin.

We post-processed the recordings analogically to those from the reading session. We again split MOV files into mp3 and mp4 ones and then aligned the audio with the text. The task was however considerably easier than with the reading sessions as we had already stored the transcriptions and temporal coordinates of the corresponding audio segments. The only necessary task was to align the start of Tali's speech with the moment Tali started to talk and the rest could have been computed from that using the information saved during conversation creation. However, we still manually inspected the segments to verify the collected data.

In conclusion, we collected a large majority of the data (for exact numbers see Chapter 5) using this method as it was engaging for Tali, fast to process and as representative of the real-world conversations as we were able to get.

4.1.3 Recording day

When discussing Tali's case with professional phoneticians, we were recommended to make some recordings also in a professional studio that would guarantee low SNR. For this purpose we organized a recording place that took place in The Recording studio of Nad Orlí Theater in Brno.

During the recording day, we made multiple recording sessions of different structures. Most notably, we captured around ninety minutes of real-world conversations. Furthermore, we obtained 7 recordings of Tali singing her favourite songs.

We recorded only audio during the recording day. Whenever Tali started to speak, we saved the timestamp using a simple dedicated program. For real-world conversations, we also transcribed her speech in real time. This was only possible thanks to the presence of Tali's mother who translated most of her words and thanks to two assistants who had some experience with Tali and helped with transcription, time-stamping and keeping the conversations engaging.

In postprocessing we aligned the first moment of Tali's speech with the first timestamp. We then manually aligned audio segments around those timestamps with the transcriptions taken during the recording day.

5 Data Analysis

Our design choices were largely motivated by the specifics of Tali's voice. We find it therefore appropriate to mention those as an explanation and justification of some techniques described in Chapter 6.

5.1 Qualiative Data Analysis

After listening to several minutes of Tali's speech and looking and corresponding spectrograms, several irregularities come out clearly. We illustrate them in the plot below.

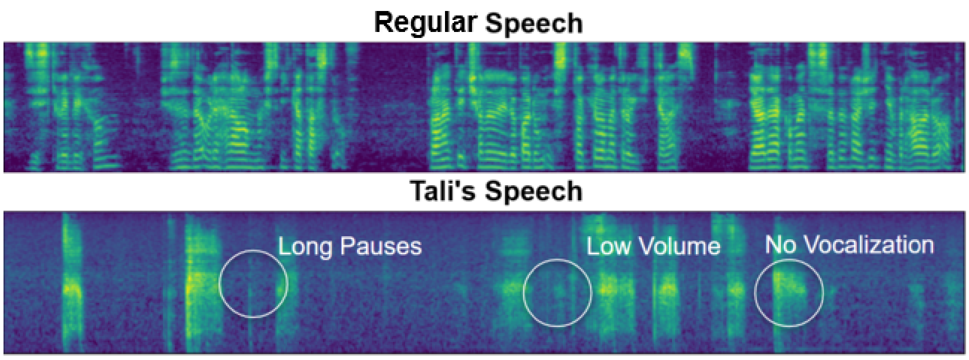

Figure 5.1: 128 Mel spectrograms of regular and Tali's speech

1. **Absence of vocalization**: The harmonic component of the speech is missing. In the plots above, this is demonstrated by the absence of parallel horizontal stripes that are prominent in the plot of the regular speech. However, unlike in the provided plot, Tali was sometimes vocalizing the first vowels of words, especially if they occurred at the beginning of sentences and the respective sessions.

2. **Long pauses**: Tali's speech is slow by itself but her rate of speech is further dampened by long pauses between and, even worse, within individual words. These pauses can be as long as one second (as is the case in the plot above). They represent a great challenge for automatic word detection.

3. **Low volume**: The somewhat lower contrast of Tali's spectrogram in comparison to the spectrogram of healthy speech indicates the weakness of her voice. This is indeed a challenge for our system because everyday noises such as a car driving on the street usually drown Tali out (an alternative example: the verification of data annotation as described in Section 4.1.1 lead Tali's mother to move their turtle terrarium as she found that the turtle's trampling significantly interfered with Tali's speech).

Other notable Tali's speech defects encompass a tendency to sometimes repeat vowels or a slight voice tremble. However, we have identified the above as the most serious.

5.1.1 Expert Qualitative Analysis

We have also reached out to expert phoneticians (As. Prof. Radek Skarnitzl and Dip. Ing. Tomáš Bořil, PhD from Institue of Phonetics, Philosophical Faculty, Charles University in Prague) and asked them to analyze Tali's speech as well. Their findings mostly correspond to those of us:

1. **Periodic source of speech is missing**: This observation builds upon source-filter model of the human speech [19]. In the source-filter approximation of the speech production, we view the speech as the composition of periodic and noise signals produced in vocal folds which are modulated by linear filters implemented by oral and nasal cavities. Tali's most critical problem was found to be the absence of the periodic part. This expert finding points to the same phenomena as what we called *Absence of vocalization*.

2. **Low SNR**: The phoneticians found Tali's speech to be rather quiet and easily drowned out by surrounding noise. For this purpose they also recommended making some recordings in a dedicated recording studio.

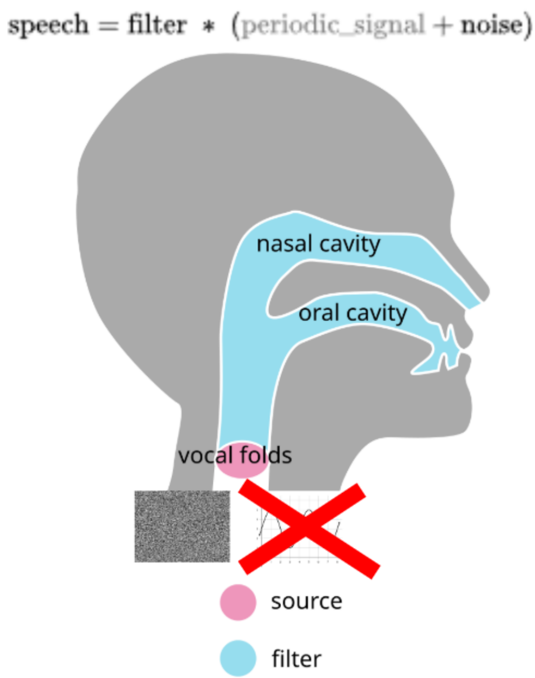

Figure 5.2: Source-filter model of speech production: we highlight the missing periodic component of Tali's speech (the image taken from Wikipedia [62])

5.2 Quantitative Data Analysis

In the table below, we provide also some quantitative attributes of our dataset. We provide these values separately for data of very collection method (i.e. *Reading sessions*,*Artificial conversations*,*Recording day*). Furthermore, we provide them for *Real-world conversations* (even though they are a subset of *Recording day*) because they play a prominent role in the evaluation of our system.

For each of these subsets, we calculate the number of recording files, total recording time (i.e. total length of video files submitted by Tali), total annotated time (i.e. the actual length of the audio used for our experiments), number of samples and number of words used in annotations. From those we also compute average words per minute (for comparison: Common Voice Czech speakers use 99 words per minute on average) and words per sample.

Furthermore, we provide relative spectral flatness [38] as a measure of how noisy the recordings are[1]. We measure this quantity in multiples of spectral flatness of a healthy voice recorded using laptop microphone.

	Reading Sessions	Artificial Conversations	Recording Day	Real-world Conversations
Number of Recordings	21	69	12	2
Total Recorded Time (h)	4:08	38:13	3:00	1:31
Annotated Time (h)	3:06	20:21	1:48	0:32
Number of Samples	2 719	8 428	781	327
Number of Words	4 739	31 499	3 273	1 159
Words per Minute	25.4	25.8	30.4	32.2
Words per Sample	1.7	3.7	4.2	3.5
Relative Spectral Flatness	1.33	1.3	8.1	1.6

Table 5.1: Overview of the recorded audio dataset. The times are provided in *hours:minutes* format.

We provide most of this data as a public dataset via Zenodo[2]. We excluded the segments clearly mentioning Tali's associates.

[1]Spectral flatness is computed for every time step of a spectrogram as $\frac{\text{geometric mean of frequency bins}}{\text{arithmetic mean of frequency bins}}$. We aggregate this value for every sample using arithmetic mean and then again for all the samples also using arithmetic mean

[2]https://zenodo.org/records/15225595

6 Training pipeline

We use Whisper by OpenAI [50] as it is a state-of-the-art architecture that is also computationally feasible to train (at least in the smaller versions) and it is also well supported by a dedicated Python library [5] and plenty of pre-trained and fine-tuned versions on the internet [40, 58, 45], that can be used as a sanity check of our own setup.

On the high level, our training pipeline consists of the following steps:

1. Pre-train Whisper on regular Czech speech.
2. Transform regular Czech speech into quasi-tracheostomy speech.
3. Fine-tune result of step 1 on this quasi-tracheostomy speech.
4. Finally, fine-tune result of step 3 on Tali's speech.

Below, we provide details on the less trivial techniques used for training. The exact hyperparameter settings for Whisper experiments and their results can be found in Chapter 7.

6.1 Pre-Training

As collecting and labelling recordings from Tali required considerable effort both from us and from her family, we decided to pre-train our models on regular Czech speech and also simulated tracheostomy speech.

6.1.1 Pre-Training on Regular Czech Speech

Default Whisper models as provided by OpenAI [5] are multilingual and they have been trained also on Czech speech, however, especially when of smaller sizes, they perform suboptimally and benefit from being fine-tuned.

Although, there are multiple fine-tuned Whisper models on Hugging Face [40, 58, 45] we fine-tuned them ourselves which was a useful sanity check of whether our Whisper training loop is sane.

We used Common Voice CS 19.0 [8, 25] for the fine-tuning. This dataset provided us with 5.5 GB recordings of 266 hours of speech. 78 of those 266 hours were verified. Both collection and verification were crowd-sourced by 1 023 speakers. In a questionnaire for the speakers, 54% of them filled in to be male while only 22% to be female. Furhtermore, 47% of speakers filled in to be 30 - 39 years old, another 18% filled in age 20 - 20. Only 2% contributors filled in an age under twenty, which is the group Tali belongs to.

6.1.2 Pre-Training on Quasi-Tracheostomy Speech

Furthermore, we attempt to simulate Tali's impediments and turn regular Czech speech into a quasi-tracheostomy one. We formalize this problem using the following formula:

$$\arg\max_{t} \operatorname{sim}(t(X_{\text{regular}}), X_{Tali}) \tag{6.1}$$

Where t is a transformation function and sim a similarity measure between transformed regular speech and Tali's speech.

Our similarity function receives two sets of audio files and then computes the score as follows:

1. Computes 512-dimensional voice embeddings of all the audios using voice embedding by pyannoteAI [15].
2. Clusters the data to two clusters with K-Means.

3. For each cluster, it computes its inner entropy of original classes (i.e. quasi-tracheostomy and tracheostomy).
4. Similarity is then computed as a mean of these two entropies.

Note that this similarity achieves its minimum when even after transformation, the voice embeddings are so distinct that K-means separates them perfectly which would lead to entropy 0[1]. On the other hand, if K-means completely fails to copy the original classes, entropy will be 1[2].

By manual experimenting and local hill-climbing that refined its results, we found the following combination of transformations as the most helpful:

- **Removal of the periodic voice component**: Inspired by the phoneticians' insights on missing periodic sources in the source-filter model, we removed it also from the regular speech. For this purpose, we utilized a code they provided us with, that makes use of linear predictive coding to separate the periodic component and replace it with white noise.
- **Time stretching**: we slow down the regular speech by a factor of four. This number was chosen to reach Tali's rate of speech which is indeed around four times slower than that of regular speech (25-30 words per minute for Tali vs 100 - 150 words per minute for regular speech).
- **Pause insertion**: To simulate Tali's tendency to intertwine her speech with pauses, we inserted seven silent segments of length 250 ms to randomly sampled timepoints of the recordings.

We tested effectiveness of these preprocessing steps on 60 one-sentence-long recordings of regular speech and another 60 recordings of the Tali's speech. Before any transformations were applied, the intra-cluster entropy was 0.335. After the transformation steps, it was 0.781 on average.

[1] $\mathbb{E}[-\log_2 X_i] = -\sum p(X_i)\log_2 p(X_i) = -1\log_2 1 = \mathbf{0}$
[2] $\mathbb{E}[-\log_2 X_i] = -\sum p(X_i)\log_2 p(X_i) = -(0.5(-1) + 0.5(-1)) = \mathbf{1}$

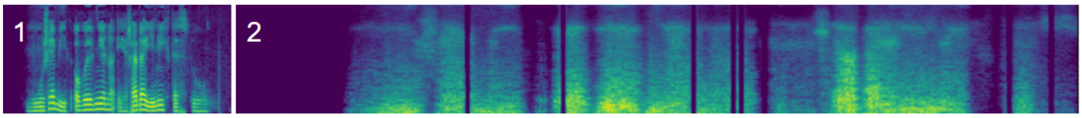

Figure 6.1: 120 Mel spectrogram of a regular voice before (1) and after (2) the transformations.

6.2 Knowledge Distillation from MLM

After inspecting the transcriptions by Whisper Tiny, we have found that even when provided with ground truths of all previous tokens when predicting the next one, it often predicts words that not only do not exist in Czech vocabulary but would be also hardly pronouncable. In order to mitigate this issue, we tried to improve Whisper's understanding of language by knowledge distillation from a MLM[3].

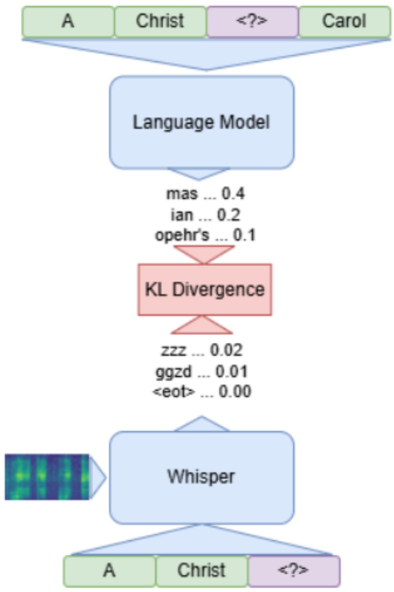

Figure 6.2: Schematics of KD from MLM to Whisper

For our language model we have used a neural network consisting of embedding layer, 6 bidirectional LSTM layers [13] and layer normalization [10] (60M parameters). We used

[3]Masked language models (MLMs) are language models that take a sentence with one more tokens being masked out and predict what the original value of the masked tokens.

the same byte pair encoding [26] tokenization as Whisper. As the training training data we used CC100-Czech Dataset [22] that contains 4.4GB of Czech text data crwaled from the internet. Due to our limited compute power, we however used only a subset of 210 MB and approximately 1.3 millions of sentences. We trained the bi-LSTM model to predict a masked tokens of these sentences. Our model achieved 93% top-5 validation accuracy after 29 epochs.

We then included this model to our Whisper pipeline as a KD teacher. For every transcription token $\mathbf{t}_t$ the loss was defined as:

$$L_t(\mathrm{asr}, \mathrm{mlm}, \mathbf{t}_{1:T}, \mathbf{s}) = \lambda_{lm} CE(\mathrm{asr}(\mathbf{t}_{1:t-1}, \mathbf{s}), \mathbf{t}_t) + (1 - \lambda_{lm}) KLD(\mathrm{asr}(\mathbf{t}_{1:t-1}, \mathbf{s}), \mathrm{mlm}(\mathbf{t}_{1:t-1 \cdot t+1:T})) \tag{6.2}$$

Where asr stands for Whisper model, mlm for the MLM teacher, CE for cross-entropy loss, KLD for Kullback-Leibler divergence loss, $\mathbf{t}_{i:j}$ for sequence of i-th to j-th transcription tokens ($\mathbf{t}_{1:T}$ for the complete transcription) and $\mathbf{s}$ for the spectrogram of the recordings.

We used KD from MLM for pre-training on both regular and quasi-tracheostomy Czech speech and for the downstream task.

Below, we provide an example of transcriptions with and without KD from MLM. Although there are plenty of errors in both, this figure already demonstrates a slight improvement

Sample 1

R: Na této silnici jsou dvě autobusové zastávky.

O1: V této silnici jsou dvě od mousové veávky.

O2: Na této vněi jsou dvě od nemusléé zastávky

Sample 2

R: Což ale platí lidem originálnosť!

O1: Což ale platí lidem originál..?

O2: Což ale platí lidem originální.?

Sample 3

R: Aféra vyvolala řadu demonstrací po městě a musela zasahovat policie.

O1: Aér vyvolala př adu drí o těstě a musela zaáovat policie.

O2: Agh titvolala př�adu devrací po městě a musela zachovat policie.

Sample 4

R: Dalších dvacet mužů bylo vážně zraněno.

O1: Dalšíím vvaaceti mu žů bylo vášné.raně.

O2: Dalšíím dvaceti mžů bylo vášně v raděno

Sample 5

R: Písmena jsou v takovém případě dvakrát vyšší.

O1: Jiímena jsou de krové vípaděpakvakrát vyšší.

O2: Tímena jsou tak takovém případě ivarát vyšší.

Sample 6

R: Architektem byl Percy Allan.

O1: Architektem byl pen Six Suite

O2: Architektem byl pes Al.

Figure 6.3: Transcriptions with and without KD from MLM. For every sample, **R** stands for the true transcription, **O1** for transcription without KD and **O2** for transcriptions with KD. The non-existing words are marked by red and the very apparent misfits by green.

Note that these transcriptions were produced in a validation run of respective Whisper training loops and every token is therefore predicted using the knowledge of its true predecessors which is fundamentally different from the actual real-world transcription where the textual context is formed of its previous predictions.

7 Experiments

In this chapter, we provide the exact hyperparameter settings and the results of our experiments with Whisper. We focus on the following three stages:

1. Regular Czech speech
2. Quasi-tracheostomy speech
3. Tali's speech

We provide results for models of three sizes - Whisper Tiny (39M parameters), Whisper Base (74M parameters) and Whisper Small (244M parameters). Furthermore, we distinguish two pipelines - Baseline and Adapted. The Baseline does not use KD from MLM and pre-trains Whisper only on regular speech. Adapted pipeline uses KD from MLM and pre-trains Whisper also on the quasi-tracheostomy speech. Due to our limited computational power, we have not trained Baseline model for Whisper Small.

Across the experiments, we have adjusted the learning rate, batch size and λ_{lm}. Other hyperparameters remained fixed for models of all sizes and for both pipelines. In all setups we used weight decay 0.01 (biases and layer normalization weights were excluded from the decay). For the KD experiments, we used temperature 2. Also all experiments use AdamW [43] optimizer with epsilon 1e-8 and with linear rate scheduler with 11 warmup steps. Finally, we train all models for 14 epochs.

7.1 Regular Czech Speech

7.1.1 Hyperparameters

When pre-training on Common Voice [8] we use the following hyperparameters:

Model	Learning Rate	KD Lambda	Batch Size
Tiny Baseline	5e-4	-	8
Tiny Adapted	1e-4	1e-3	8
Base Baseline	3e-5	-	4
Base Adapted	3e-5	1e-3	8
Small Adapted	3e-5	1e-3	8

Table 7.1: Hyperparameter settings for regular Czech speech experiments

7.1.2 Results

With these hyperparameters, we achieved the results presented in the table below.

Model	Validation Loss	WER	CER
Tiny Baseline	1.236	0.447	0.031
Tiny Adapted	0.636	0.345	0.023
Base Baseline	0.819	0.303	0.019
Base Adapted	0.516	0.312	**0.018**
Small Adapted	**0.344**	**0.209**	0.024

Table 7.2: Results for regular Czech speech experiments

Note that all these results are based on single experiments and that reproducing our experiments might lead to somewhat different numbers. However, we find the results to be different enough to conclude that scaling up the Whisper improves the performance. Furthermore, KD from MLM improves Whisper Tiny significantly.

We provide our improved Whisper Tiny for regular Czech speech also to the general public via Hugging Face[1].

7.2 Quasi-Tracheostomy Speech

The subsequent pre-training on quasi-tracheostomy data was applied only for the Adapted pipeline, therefore, we leave the Baseline out of the tables.

[1] https://huggingface.co/Hobit2002/whisper_tiny_cs

By quasi-tracheostomy we understand the regular Czech speech from Common Voice to which we applied preprocessing steps described in Subsection 6.1.2.

7.2.1 Hyperparameters

Model	Learning Rate	KD Lambda	Batch Size
Tiny Adapted	1e-4	5e-2	8
Base Adapted	3e-5	5e-2	8
Small Adapted	1e-5	1e-3	4

Table 7.3: Hyperparameter settings for regular quasi-tracheostomy speech experiments

7.2.2 Results

Model	Validation Loss	WER	CER
Tiny Adapted	1.596	0.627	0.049
Base Adapted	1.273	0.556	**0.041**
Small Adapted	**1.000**	**0.488**	0.042

Table 7.4: Results for quasi-tracheostomy experiments

7.3 Tali's Speech

Finally, we provide our settings and results for the experiments on Tali's speech.

7.3.1 Data Split

We use traditional train, validation, test set split for our experiment. We distribute data to data splits based on the type of the session by which they were collected:

- **Reading session data** is used only for training as we do not find it to be sufficiently representative of the target domain to be used for validation. The samples from reading sessions are often rather short and capturing only a few words thus providing

Whisper with unrealistically limited context. To account for this caveat, we merge neighboring samples (and the audio inbetween them) into at least five second long segments.

- **Artificial conversations** are used for training, validation and the final test as it is the dominant type of data in our data set.
- **Real-world conversations** collected on the recording day are used only for the final test.
- **Other recording day data** (singing and reading pre-scribed conversation) is used only for training as we do find it representative of the target domain.

The resulting distribution of data in our data split is depicted in the table below. We differentiate the test data into *In-Domain* which consists of artificial conversations and *Out-of-Domain* that consists of artificial conversations.

Data Split	Sessions	Samples	Time
Train	74	8 374	16:31
Validation	11	1 418	3:14
Test: In-Domain	5	682	1:56
Test: Out-of-Domain	2	327	0:32

Table 7.5: Data split for experiments with Tali's speech

Note that session-wise and sample-wise, our train:validation:ration roughly corresponds to common 8:1:1 split.

7.3.2 Hyperparameters

Model	Learning Rate	KD Lambda	Batch Size
Tiny Baseline	1e-4	-	8
Tiny Adapted	1e-4	1e-3	8
Base Baseline	3e-5	-	8
Base Adapted	3e-5	1e-3	8
Small Adapted	1e-5	1e-3	4

Table 7.6: Hyperparameter settings for regular Czech speech experiments

7.3.3 Results

Finally, we provide the results which we obtained on the final test set. Before the evaluation, the models were trained on both train and validation sets. We state the results for *In-Domain* and *Out-of-Domain* data separately.

Model	Validation Loss		WER		CER	
	In	Out-Of	In	Out-Of	In	Out-Of
Tiny Baseline	0.931	3.279	0.452	0.975	0.059	0.111
Tiny Adapted	0.689	2.376	0.402	0.854	0.037	0.080
Base Baseline	0.681	2.417	0.361	0.831	0.032	0.061
Base Adapted	**0.569**	2.027	0.349	0.817	0.029	**0.063**
Small Adapted	0.574	**1.995**	**0.319**	**0.774**	**0.0284**	0.068

Table 7.7: Performance of models on different metrics in in-domain and out-of-domain settings.

Note that the validation WER and CER are computed in a rather misleading way because even during validation every token is predicted with the true knowledge of the previous tokens [2]. On the other hand, for the real-world transcription, the labels are not available and every token is therefore predicted using the previous predictions. Thus, based on these validation results we cannot make strong conclusions of the actual transcription quality.

[2] This seems to be a common practice among the tutorials we found for Whisper fine-tuning [35, 29]. It has been apparently also used by the authors who published Whisper Tiny, Base and Small fine-tuned on Czech Common Voice on Hugging Face [40, 58, 45] as we replicated their WER scores up to 5% this way.

7.4 Discussion

Even with the aforementioned caveat in mind, there are several insights we can draw from these results: there is a huge difference between the performance for artificial conversations (*In-Domian*) and the real-world conversations (*Out-of-Domian*). The magnitude of the difference comes to us surprising. Clearly, recordings of the real-world conversations had their specifics: the audio segments do not capture the whole sentences but rather groups of 2-3 words and relatively often, the ending vowels of the words are cut out. These complications were hard to avoid because Tali's mother, who translated to us her speech, did not usually wait to the very end of the sentences (and sometimes not even to the very end of the words themselves). Cutting out the mother's voice, which we deemed necessary, then forced us to limit the context provided by individual samples.

For *In-Domain* Whisper Tiny and Base achieved WERs and losses comparable to those for regular speech, with adapted models pipelines achieving considerable yet not drastically better results. The best results were achieved by Whisper Small, however, they are significantly worse than those achieved for regular speech. Whether the problem lies in our adaptations not working well for Whisper Small, in the limited information content of the data or if it has some other cause, remains to be an open question.

To sum up, we find it highly likely that in order to work well, the Whisper will have to learn from the data collected during the deployment. Fortunately, the results for artificial conversations give us hope, that when provided with enough data from the target domain, the Whisper will achieve good performance.

8 Deployment

When transcribing Tali's standard speech, our models tend to hallucinate, possibly beacause of her low rate of speech. However, when we asked her to make longer pauses between her words (a habit recommended also by her therapists) and then squeezed the silent segments in these recordings to 500ms, we achieved the results below.

True transcription	**Predicted transcription**
Chápu naučím vás zemské etiketě	chápu naučím nás lidské endemie
To jsou pravidla slušnosti	To jsou pravidla a slušností
Začněme s podáním ruky	Začneme podáním luky
Stoupnete si usmějete se	stoupnit. Usmějte se.
Ruku podáváte ze předu	luku po tvrdém řadu předu.

Table 8.1: Comparison of true and predicted transcriptions with word-level coloring: the green words perfectly match the ground truth and the orange are close to it.

We find these transcriptions reasonable and so does the Tali and her family (see their feedback in Figure 8.1). However, in order to provide the users with intermediate responses (i.e. partial transcription before the whole sentence was finished) and to alleviate high inaccuracy by displaying multiple possible transcriptions, we furhter experiment with word-by-word transctiptions.

In order to estimate our deployment performance, we collected 3 artificial conversation sessions where Tali was asked to make at least one second long pauses inbetween her words. These sessions were then used to experiment with different deployment strategies. We have not used these recordings for Whisper training experiments.

Once collected and annotated, we split each of the lines by Tali to shorter segments containing exactly one word. We then compute top five transcriptions for each of segments of each line and measure the relative frequency of correct transcriptions.

We experiment with the following strategies. In all cases we use beam search of 15:

- **Unguided beam search transcription**: we apply the standard beam search transcription to obtain the five most likely transcriptions. We transcribe every word segment independently from the previous ones.
- **Guided beam search transcription**: we apply the standard beam search transcription, furthermore we constraint the possible transcriptions only to all existing Czech words (as provided by [6]). We store this vocabulary in a prefix tree [16]. When predicting a new token, we mask out all logits except for those stored in the prefix tree as possible suffixes of the previous tokens. We transcribe every word segment independently from the previous ones.
- **Guided beam search transcription with merged segments**: we, again, apply the the standard beam search and use the prexif tree with Czech vocabulary. Furthermore, we merge every word audio segment with the previous ones and also provide the model with the true transcriptions of the previous words.

These strategies lead to following results:

Strategy	Top-1 Accuracy	Top-2 Accuracy	Top-5 Accuracy	Time per word (s)
Unguided	0.29	0.35	0.42	5.5
Guided	0.30	0.34	0.40	**4.8**
Guided + Merged	**0.36**	**0.41**	**0.46**	4.9

Table 8.2: Results for different decoding strategies

The Tali and her family confirmed to us that the results achieved by direct transcription with dropped silent segments and *guided beam search transcription with merged segment* would be already helpful to them (especially for communication with Tali's assistent in school and with her therapist). Overall, they provided us with the following feedback:

> It looks pretty good from the preview. Tali has quite learned how to separate words, so she keeps to it when speaking, which helps everything. Another side effect of the recording is that she strengthens her voice by recording. Even to the point where it's noticeable. So overall, our efforts are beneficial on many levels.

Figure 8.1: Feedback from Tali's mother (we received also several others but this is the most elaborate one)

8.1 Application Architecture

After receiving such a positive feedback, we shift our focus to the development of an application that would facilitate the system in the real-world. This is no longer part of this thesis. However, for an interested reader we provide a high-level outline of the system we aim to design:

1. The system shall be implemented as a web application accesible through web browser.
2. When Tali opens the application, microphone of her phone's microphone shall be activated and the link for the conversation partners shall be provided (in the form of a QR code).
3. Conversation partners can connect to the application as well using the link generated by Tali's application.
4. An algorithm running in the background shall sample the segments of Tali's speech from the audio stream. The exact nature of this mechanism is yet to be decided. At the moment, we plan to use the voice embedding by pyannoteAI [15] to decide whether the incoming segments of audio contain Tali's speech or not using similarity measures.
5. Another, computationally powerful, device shall connect to the web server as a *transcribing client*.
6. The segments with Tali's speech shall be sent to the server where they are forwarded to the *transcribing client*.
7. *Transcribing client* shall return the most likely transcriptions and the web application shall display them both to Tali and her conversation partners.
8. Conversation partners can select the correct transcription. If they do so, ASR system shall use it as a context for the later segments.
9. Alternatively, conversation partners can indicate that the provided transcriptions were bad and that Tali should repeat the respective word.
10. The feedbacks from the above points shall be used to further train the system.

11. The web application shall also provide an interface to create a purpose-specific vocabularies in order to increase the performance for purpose-specific conversations. However, the exact formation of these vocabularies is still an open problem.

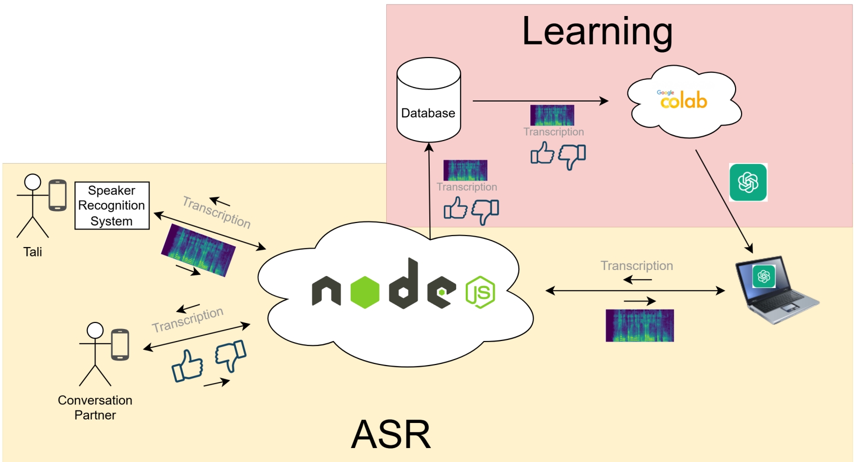

Figure 8.2: Scheme of the proposed application.

This design is not optimal for a long term usage as it requires a transcribing client (our laptop, to start) to connect to the web server every time the application ought to be used. However, it can be easily developed with the resources that we currently have and so we find it well-suited to rapidly and cheaply deploy the prototype.

9 Conclusion

This thesis investigates development of an AAC ASR system for a patient with a severe speech impediment caused by tracheostomy and possibly also by a neurological damage.

We collect and annotate twenty four hours of her voice. To this end, we develop a novel data collection method: artificial conversations. This strategy greatly reduces the time required for the data labelling. We analyze different methods of data collection. We also create a public dataset of the Tali's speech. Interstingly, the feedback from Tali's family indicates that the artificial conversations have also positive logopedic effects.

We train Whisper models of multiple sizes on the collected data. We utilize pre-training on regular Czech speech and quasi-tracheostomy speech. We also implement KD from MLM into Whisper. These adaptations were chosen during experiments with Whisper Tiny and they, indeed, lead to major improvements for this model. Especially the KD from MLM leads to significant improvements even for regular Czech speech. However, for larger models, these adaptation seem to bring increasingly limited benefits.

As our models are not accurate enough for standard end-to-end transcription, we develop several startegies on how to use them for real-world word-by-word decoding. We reach a performance that largely surpasses that of untrained human listeners and that is considered helpful also by the patient and her family. Finally, we outline our plan for an application that would facilitate this sytem to the patient.

9.1 Future work

As already mentioned, although useable, our system still does not achieve the accuracy of Tali's family members or caretakers indicating that there is still a room for improvement. Namely, we find creating a more realistic quasio-tracheostomy voice or ensembling Whisper with lip-reading models and LLMs (for better semantical understanding) to be promising directions.

At the same time, the system would benefit from reduced computational expenses. Further improvements of Whisper Tiny performance combined with efficiency measures such as quantization might remove the need for an external server thus cutting the financial burden of our application dramatically.

Nevertheless, as our tool performs sufficiently well even in its current form, we find it reasonable to put it into practical use as soon as possible and let the real-world insights guide our research.

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